

Grinding Rollers with Taper Roller Bearings

Table of Contents:

	Sheet
1. Description of Bearings	2
2. Assembly	2
2.1 Necessary Auxiliaries	2
2.2 Preparations	3
2.3 Mounting of the Bearings	3
2.3.1	3
2.3.2	4
2.3.3	5
2.3.4	6
2.4 Alternative for Mounting of the Bearings	6
2.5 Dismounting	8
2.5.1 Auxiliaries for Dismounting	8
2.5.2 Preparations	8
2.5.3 Dismounting of the bearings	8
Picture 1	10
Picture 2 and 3	11
Picture 4	12
Picture 5	13

"Assembly Instructions for Slip Ring Sealing" see Table of Contents of
LOESCHE-Mill

1. Description of Bearings

The grinding rollers are equipped with one cylindrical roller bearing, type NU as floating bearing and a double taper roller bearing in X-arrangement as fixed bearing. The distance sleeve between the outer rings of the double taper roller bearing is tolerated by the manufacturer to achieve a suitable axial clearance for the special application. Adjustment of clearance at assembly is not necessary.

The arrangement of the bearings is shown on sheet 10, Picture 1.

Depending on the manufacturer the double taper roller bearing has 2 single inner rings or only one inner ring.

2. Assembly (see Picture 4, sheet 12)

2.1 Necessary Auxiliaries

To facilitate the assembly we recommend the following equipments and tools:

- a) lifting crane (capacity according to the weight of 1 grinding roller; see assembly drawing "Grinding roller cpl.")
- b) holding device for roller axle to enable assembly in vertical position
- c) gas ring burner (1 - 2 units) or electrical heater for heating-up of the roller hub
- d) oil bath for heating-up of the bearings. Max. oil bath size approx. 900 mm (35 in) dia. Requested oil quantity max. 150 l (40 gal).
- e) 3 ring screws DIN 580 (depending on bearing size M 12, M 16, M 20) for suspension of the outer ring of taper roller bearings
- f) lowering device for the enclosed outer ring of taper roller bearing (see Picture 2, sheet 11).

2.2 Preparations

The anti-friction bearings must be protected against dirt and humidity during mounting. Therefore, the assembly place has to be dust-free and dry. All parts to be fitted have to be absolutely clean and free of chips and castings and free from moulding sand residue.

The anti-friction bearings resp. parts of anti-friction bearings should be unpacked just before the assembly. The bearings are preserved at time of assembly and must not be washed except they become dirty during assembly under special conditions. In this case they have to be cleaned with washing petrol or benzene (fire risk!). Complete bearings have to be revolved during cleaning. Immediately after drying bearing must be protected by oil or grease against corrosion.

Each component of the anti-friction bearings - inner ring and outer ring with roller cage assembly of the cylindrical roller bearing as well as the inner ring, outer ring and distance sleeve - is separately packed. Components to fit together are marked with identical figures resp. numbers on the rings and on the packing. Correlating bearing rings are marked with additional letters on the bearing side.

We recommend to treat the bearing fits on the axle inside the hub with MoS₂ paste G Rapid filmy.

2.3 Mounting of the Bearings

2.3.1 Put roller axle in vertical position with bearing fits to the top (holding device).

Unpack inner ring of the cylindrical roller bearing and heat up to min 50°C (125°F) above ambient temperature, max. 100°C (210°F).

Note: The heating up is executed in an oil bath or on a heating plate. Do not use a torch with open flame! Only specialists (bearing fitters) should be allowed to use a torch with soft flame in emergency cases.

Treat the fit of the inner ring on the axle with MoS₂ paste G Rapid filmy.

Put inner ring into position until it touches the axle shoulder.

Put distance sleeve for inner ring of taper roller bearing into position.

Heat inner ring(s) of taper roller bearing together with roller cage assembly and distance sleeve for the outer rings up to 100°C (210°F) in an oil bath.

Put the mentioned bearing components onto the roller axle until the inner ring touches the distance sleeve.

Screw cover disc for tensioning of the inner rings onto the top face of the axle.

2.3.2 Assembly procedure described under this item must be executed without interruption to utilize the high temperature of the hub and the corresponding enlargement of the bearing fits.

Screw sealing cover to hub face (not necessary in case of hubs with cast shoulder).

Heat up the hub in the range of the bearing fits from outside to 80°C - 100°C (180°F - 210°F).

Note: If possible use 1 or 2 ring burners operating to the center. The time for heating up can be reduced with the use of 2 burners. If no ring burner should be available a normal torch can be applied for equal heating up with a soft flame.

A better method is the use of an electric heater inside the hub bore. The hub has to be covered by a tent square. Since heating up takes several hours this procedure should take place during the night if sufficient time is available.

Put hub into vertical position with sealing cover or shoulder upside down.

Do not support cover directly on the ground.

The hub should be positioned at a height that allows free projection of the roller axle after insertion from above when the bearings are in correct position.

Threat bearing fits inside the hub with MoS₂ paste "G" Rapid or equivalent.

Insert outer ring of cylindrical roller bearing together with roller cage assembly until touching the sealing cover resp. the hub shoulder.

Insert distance sleeve of the hub until touching the outer ring of the cylindrical roller bearing.

Put outer ring of taper roller bearing into position using a lowering device.

2.3.3 Insert grinding axle together with pre-mounted bearing components vertically from top into the hub.

Rotate the roller axle during inserting the inner ring of the cylindrical roller bearing into the roller cage assembly in order to avoid scuff marks.

Rotate axle again as soon as it is supported by the taper roller bearing. This action is very important to achieve correct positions of the taper rollers and to avoid oblique setting of the rollers.

Insert front ring of the taper roller bearing into the hub until it touches the distance sleeve of the bearing outer rings. Rotate axle again.

Screw hub cover together with O-ring sealing to the hub during rotating the axle and clamp bearing outer ring axially.

Retighten cover screws several times during cooling of the hub using a torque wrench and rotate roller axle simultaneously.

Retighten screws of the sealing cover after cooling of the readily assembled grinding roller using a torque wrench. Hub cover has to be removed during the procedure (not necessary in case of hubs with a cast shoulder).

Remount hub cover again during rotating the axle.

2.3.4 Put grinding roller into horizontal position after correct tightening of all screws and support roller.

Shrink locating ring with pin screws for taking-up the springs for the slip ring sealing onto the grinding axle.

Mount slip ring sealing.

2.4 Alternative for Mounting of the Bearings (see Picture 5, sheet **13**)

Alternatively the rollers can be assembled in a different sequence:

Support the hub at a height that allows to insert the axle in vertical position from top or from below.

Mount only the inner ring of the cylindrical roller bearing and the inner distance sleeve onto the axle.

Mount only the slip-ring sealing including the shrunk ring as far as these components belong to the axle.

Screw the sealing cover against the hub.

Heat up the hub to a temperature of 80°C - 100°C (180°F - 210°F). Using the above mentioned procedure.

Insert the outer ring and the roller cage assembly of the cylindrical roller bearing to the sealing cover respectively the hub collar.

Insert outer distance sleeve between cylindrical and taper roller bearing into the hub.

Insert the complete taper roller bearing into the hub. The inside position outer ring of the bearing must touch the outer distance sleeve.

Brace the bearings inside the hub axially using clamps.

Oil the inner ring of the cylindrical roller bearing on the axle.

Pull the axle from below into the hub using a crane.

Rotate the axle during inserting the inner ring into the cage of the cylindrical roller bearing in order to avoid scuff marks.

Rotate axle during inserting it into the inner ring (S) of the double taper roller bearing.

Screw cover disc onto the axle face for bracing the inner rings.

Screw hub cover onto the hub during rotating the axle.

Tighten the screws crosswise using a torque wrench (see assembly drawing "Roller cpl.").

The mounting of the bearings may not be done without rotating the axle because otherwise a too small axial clearance or zero-clearance might arise.

2.5 Dismounting

2.5.1 Auxiliaries for Dismounting

We recommend the following equipment and tools for dismounting:

- a) crane
- b) dismounting device according to Pictures 2 and 3 on sheet **11**
- c) hydraulic jack with a travel of approx. 70 - 100 mm (2.8 - 4 in) and a force of about 1000 N (225 lbf)
- d) hydraulic pump, manually operated, resp. high pressure pump of the LOESCHE tool box for the fluid injection between the mating surfaces.

2.5.2 Preparations

Dismount hub cover, cover disc and - if existing - the sealing cover.

The following dismounting procedure should take place at vertical position of the roller. The hub will be stripped off the axle to the top.

2.5.3 Dismounting of the bearings

Dismount the hub cover.

Loosen the inner ring(s) of the taper roller bearing on the axle:

Unscrew the plug on the axle face; pump high pressure oil (viscosity 114 cSt (50°C) = 1227 * 10⁻⁶ ft²/s (122°F)) into the grooves of the bearing seats between the inner ring(s) and the axle.

The inner ring(s) will expand under the operation of the high pressure oil and the bearings can be stripped off the axle without difficulty.

Pull off the outer ring of the taper roller bearing on the cover side.

Please note: The outer ring is provided with threaded holes for ring screws (DIN 580).

With hubs made of cast steel or meehanite the bearing fit can be untightened by using high pressure fluid injection. A manually operated high pressure pump is part of the mill tools.

For other cases we recommend to use CO₂ - snow sprayed onto the bearing ring. Due to the extremely low temperature the bearing ring contracts and becomes loose inside the hub.

Pull off the second outer ring of the taper roller bearing. Use an auxiliary equipment consisting of a channel star with movable latches.

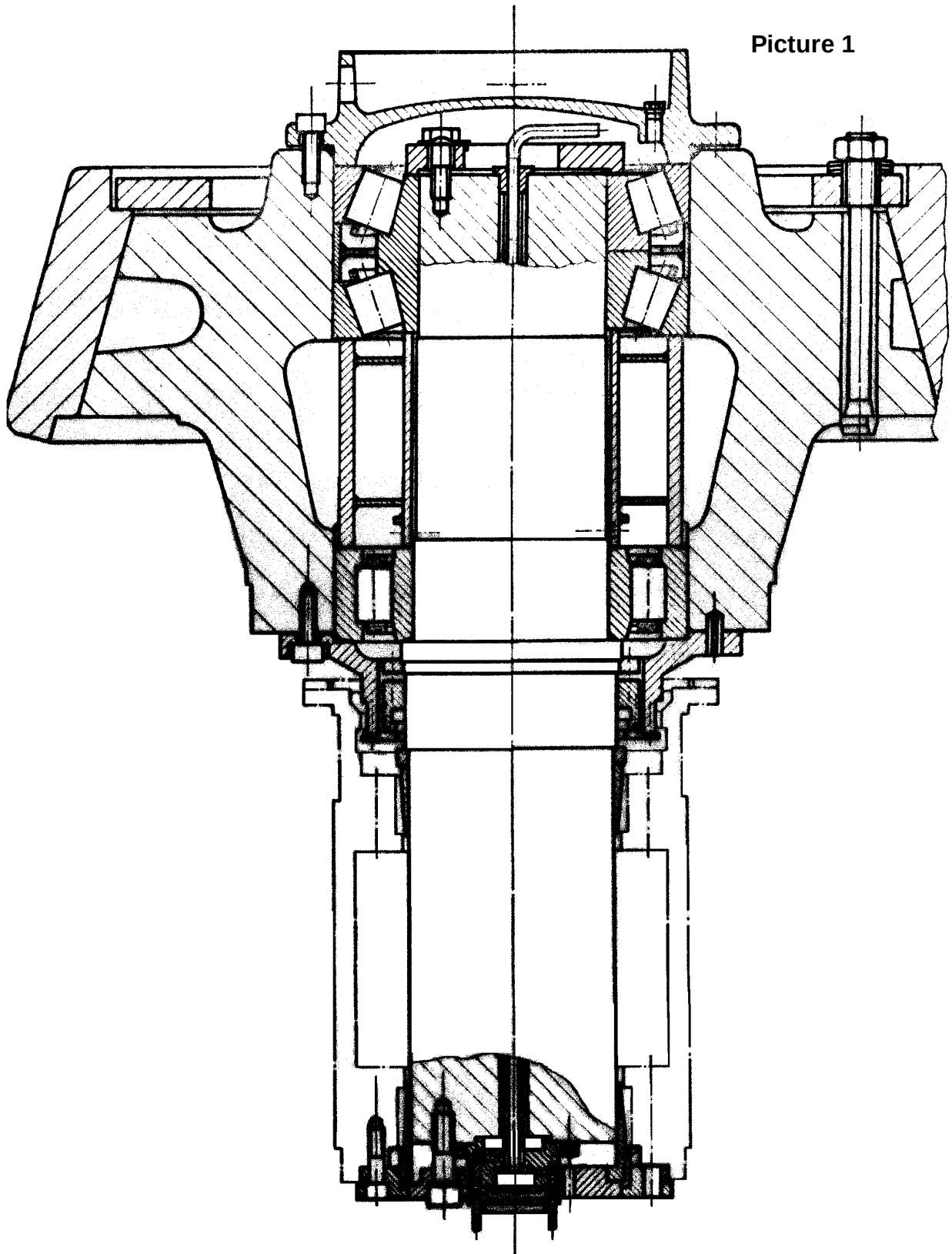
Dismount the distance sleeve between the outer rings of the taper roller and cylindrical roller bearing.

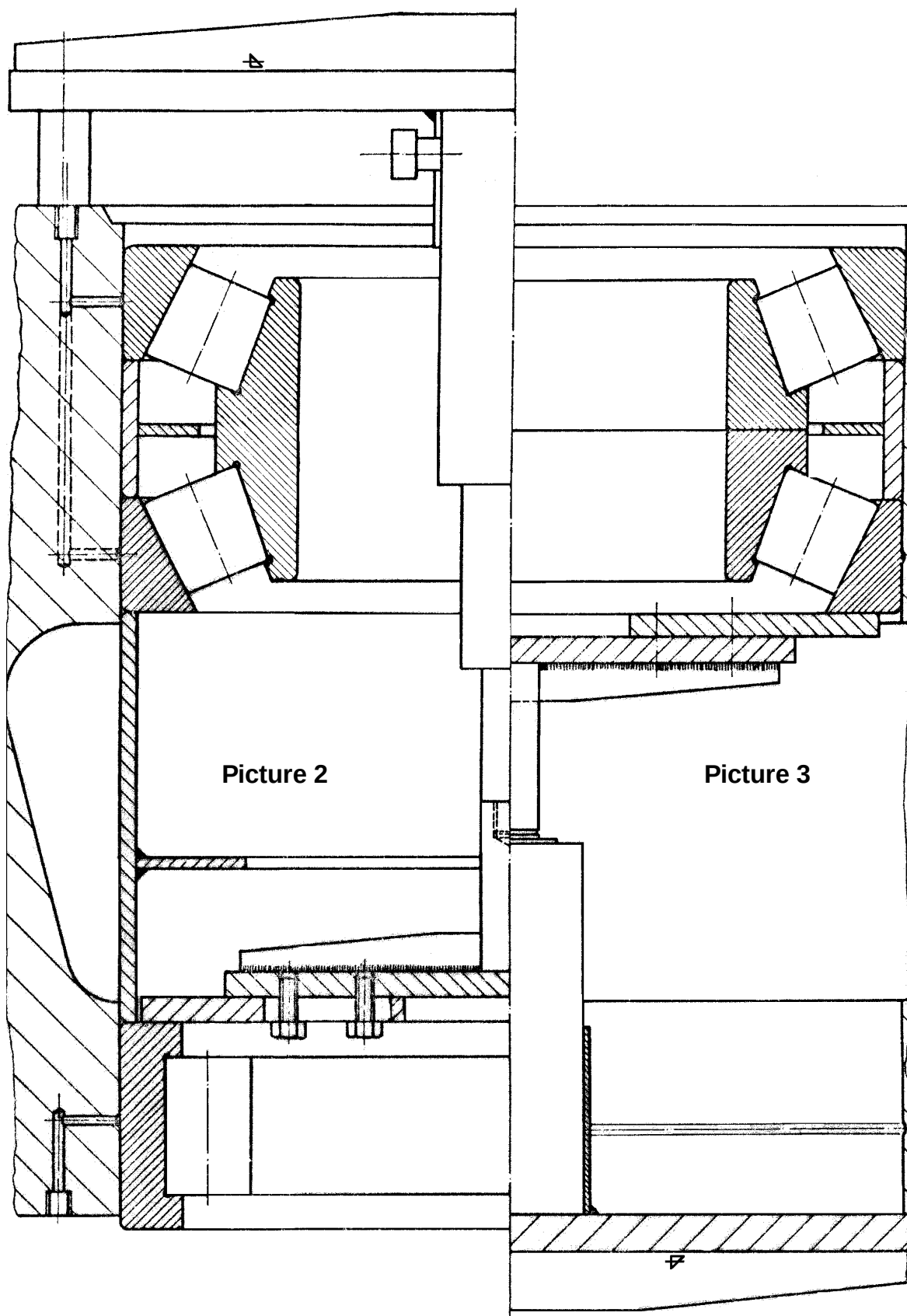
Dismount the outer ring of the cylindrical roller bearing together with the roller cage assembly.

Please note: Cylindrical roller bearings inside a roller with a screwed sealing cover can be removed by applying an hydraulic jack. This procedure can be facilitated by using CO₂ - snow. (See Picture 3, sheet **11**).

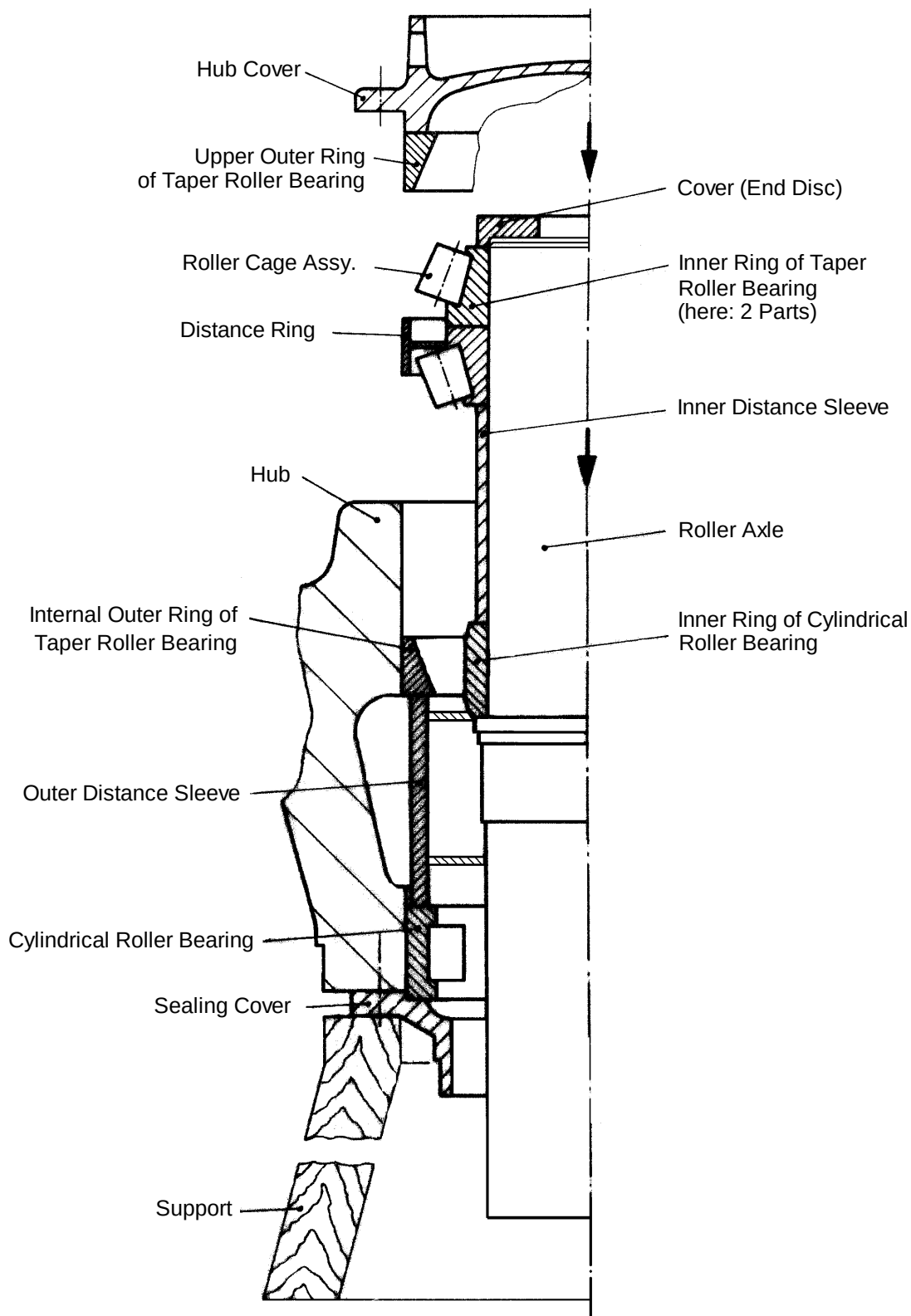
The CO₂ - snow method should be applied in any case if the roller hub is equipped with a cast shoulder in the range of the cylindrical roller bearing. Should CO₂ - snow not be available, the hub can be heated-up in the range of the cylindrical roller bearing at simultaneous assistance of pressing-off screws for the outer ring of the cylindrical roller bearing inside the cast shoulder of the hub.

Strip off the inner ring of the cylindrical roller bearing from the axle after heating it up with a soft flame (this procedure is only necessary if the bearing ring must be replaced).





Picture 4



Picture 5